

Normal measures from critical-sequence tail classes

A canonical Prikry layer at the ultraexacting boundary

Prepared for Vladimir Reshetnikov

18 September 2026

Abstract. Let λ be ultraexacting. We construct a class q of cofinal ω -subsets of λ modulo finite difference with the following property. For every parameter $p \in V_\lambda$, the inner model

$$M_{q,p} = \{x : \text{tc}(\{x\}) \subseteq \text{OD}(q, p)\}$$

contains a normal measure on λ , defined externally by eventual membership along q . Every representative of q is Prikry generic over $M_{q,p}$ for this measure. The same q works for all such p . The key new step is a regressive-function argument at the *first* critical point, combined with the archive's minimal-parameter-rank method.

The resulting consistency statements

A definability obstruction. Ultraexactingness contradicts the principle that every cofinal tail class admits a splitting subset of λ ordinal definable from that class and a parameter in V_λ .

An enhanced equiconsistency package. The theory asserting an ultraexacting λ with $V_\lambda \subseteq \text{HOD}$, together with the canonical normal measure and Prikry layer constructed here, is equiconsistent with $\text{ZFC} + \text{I0}$. The underlying ultraexacting-I0 comparison and the coding step are imported; the additional canonical structure is proved here.

A stationary-set obstruction. A strongly compact cardinal below λ prevents the canonical Prikry layer from being the entire universe. Any further outer extension satisfying that hypothesis must collapse its successor of λ or destroy a specific stationary set. The stationary mechanism is the archive's earlier one, now applied to the newly constructed layer.

Scope and status. This is a conventional, unrefereed research argument, not a claim of an unconditional inconsistency of ultraexacting cardinals or I0. All new implications are proved in English. No new Lean verification is claimed. The measure is complete *inside the indicated inner model*, not in the ambient universe, where $\text{cf}(\lambda) = \omega$. The contribution is a continuation of the supplied archive; worldwide priority is not asserted.

Contents

1	What is added to the archive	3
2	Parameters, tail classes, and inner models	4
3	One witness, uniformly correct for all parameters	5
4	The two tail theorems	7
5	The internal normal-measure theorem	8
6	Canonical Prikry layers	10
7	Sharpness and diagnostic counterexamples	11
8	Conditional inconsistency and equiconsistency	12
9	The stationary-set gap above the canonical layer	14
10	Audit, limitations, and next mathematical questions	16
A	Prikry genericity and preservation: proofs	17
	References	21

1 What is added to the archive

The supplied Cardinals3.zip contains the second-edition *Large Cardinals Synthesis*, the *Unified Report*, and Lean sources. The synthesis already proves the completeness barriers, sharp-width theorems, and rigidity of certain classes in $D_\lambda/\equiv^*$; it also contains an I_0 -calibrated coding package and a successor-stationarity obstruction [1, 2]. Those results are not being counted again as new results here.

The decisive change of question is from “*Can a tail class define a small family of representatives?*” to “*Which subsets of λ contain almost all of a critical sequence, and do they form a normal measure in an inner model?*” Rigidity alone gives no answer to the normality question. Indeed, Section 7 gives an explicit recoding which retains the same relative-definability model but makes the corresponding tail ultrafilter nonnormal.

Issue	Archive baseline	Extension here
Critical-sequence class	Fixed modulo finite difference; relative rigidity	No relative-definable splitter; every relative-definable regressive function is eventually constant
Inner model at λ	Regularity, and inaccessibility in the coded boundary case	An actual <i>internal normal measure</i> in each fixed-parameter relative HOD
Prikry phenomenon	An iteration-based consistency construction	An intrinsic Prikry layer inside <i>every</i> ultraextending universe, with no iterability assumption added
Consistency strength	Ultraextendingness calibrated by I_0	The canonical measure/layer and full low-rank agreement can be added without increasing that calibration
Successor obstruction	Collapse or destruction for an arbitrary regularizing inner model	A concrete canonical intermediate model and a named stationary witness

The phenomenon that critical sequences can be Prikry generic over an iteration model is classical. A related iteration-based use occurs in [4, Section 5.2]. Our assertion concerns instead a *specific hereditary ordinal-definability class formed in the ambient universe from the tail class itself*. It includes uniformity over all additional parameters of rank below λ . Neither a well-founded iteration of the given witness nor an identification of the universe as a forcing extension is assumed.

1.1 Dependency boundary

We use the local Kunen inconsistency theorem: in ZFC there is no nontrivial elementary self-embedding of $V_{\mu+2}$ [5]. For the final equiconsistency assertion we additionally use two results of Aguilera–Bagaria–Goldberg–Lücke: the equiconsistency of ultraextendingness with I_0 , and forcing which preserves an ultraextending λ while making $V_\lambda \subseteq \text{HOD}$ [4, Theorem A and Proposition 3.9]. The normal-measure construction itself does not use the consistency comparison or the coding theorem.

The genericity criterion is Mathias’s criterion [6]; Section A supplies a proof, including the required Prikry lemmas. The strong-compactness application also includes its reflection argument. Thus those applications do not hide a new unproved forcing or reflection lemma.

2 Parameters, tail classes, and inner models

Work in an ambient universe $V \models \text{ZFC}$. All unadorned ranks, cardinals, and definability classes are computed in V . For a set x , $\text{tc}(\{x\})$ is the transitive closure of the singleton $\{x\}$, so in particular it includes x .

Definition 2.1 (Ultraexactingness). An uncountable cardinal λ is *ultraexacting* if, for every sufficiently large cardinal $\zeta > \lambda$, there are

$$X < V_\zeta, \quad V_\lambda \cup \{\lambda\} \subseteq X, \quad j : X \longrightarrow V_\zeta$$

such that j is elementary, $j(\lambda) = \lambda$, $j \restriction \lambda \neq \text{id}$, and $j \restriction V_\lambda \in X$. We use the equivalent formulation with arbitrarily large heights; the usual definition quantifies over every height above λ [3, Definition 3.3 and Lemma 3.2].

The domain X is not assumed transitive. In particular, $x \in X$ does not imply $x \subseteq X$. Only the explicit inclusion $V_\lambda \subseteq X$ and elementary definable closure will be used.

Definition 2.2 (Tail quotient). For $\text{cf}(\lambda) = \omega$, set

$$D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}, \quad Q_\lambda = D_\lambda / \equiv^*,$$

where $a \equiv^* b$ means that $a \Delta b$ is finite. Write $[a] = \{b \in D_\lambda : b \equiv^* a\}$. For $q \in Q_\lambda$ and $A \subseteq \lambda$, put

$$\text{Tail}(q, A) \iff (\exists a \in q) |a \setminus A| < \omega.$$

The existential quantifier may equivalently be universal. Say that A *splits* q if both $a \cap A$ and $a \setminus A$ are infinite for one, equivalently every, $a \in q$.

Every representative has rank λ , and the full class q is an element of $V_{\lambda+2}$. Passing from a representative to its class forgets finitely many entries; it does not supply a preferred representative.

Definition 2.3 (The parameter convention). $\text{OD}(q, p)$ consists of the sets definable in V from ordinals and the *two individual set parameters* q, p . Equivalently one can use a unique definition in a sufficiently large rank segment with these parameters. Set

$$\mathcal{O}_q = \bigcup_{p \in V_\lambda} \text{OD}(q, p), \quad M_{q,p} = \{x : \text{tc}(\{x\}) \subseteq \text{OD}(q, p)\}, \quad M_q = M_{q, \emptyset}.$$

A finite tuple of members of V_λ can be coded by one member of V_λ , since λ is a limit cardinal.

A parameter is not a family of permitted parameters

In $\text{OD}(q, p)$ the class q is *one set parameter*. Its members are not individually available as parameters. Also, $M_{q,p}$ is not $L(q, p)$, and it need not contain q or p . If q were an element of this transitive inner model, all its representatives would be elements too. The construction below proves precisely that this does not happen.

Lemma 2.4 (Fixed-parameter hereditary OD). For arbitrary sets q, p , $M_{q,p}$ is a definable transitive inner model of ZFC. The class $\text{OD}(q, p)$ has a (q, p) -definable set-like well-order, and every nonempty definable subclass of it has a least member in that order.

Proof. Code a definition by a rank height, a formula number, and a finite tuple of ordinal parameters. The rank height is required to contain q, p and all the other parameters. A code is valid when its formula uniquely defines an object in that rank segment. Order codes first by a bounding ordinal for all their entries, and then by a fixed well-order of the set of codes with that bound. Every initial segment is a set, and every nonempty class of valid codes has a least code. Assign each definable object its least code. This gives the required definable set-like well-order. The satisfaction relation used here is the definable satisfaction relation of a *set* structure, not a truth predicate for V .

Write $M = M_{q,p}$. Its definition immediately gives transitivity and membership of every ordinal. Whenever an object is definable from ordinals, q, p , and finitely many members of $\text{OD}(q, p)$, replace the latter parameters by their definitions. The object is again in $\text{OD}(q, p)$. We use this substitution observation repeatedly.

For $x, y \in M$, pairing and union are ordinal definable from q, p and have only hereditarily ordinal-definable members, so their usual values belong to M . The same argument gives Infinity. For $x \in M$, the set

$$\{y \subseteq x : y \in M\}$$

exists in V , is definable from x, q, p , and has all its members in M . It therefore belongs to M and is its internal power set of x . For any fixed formula, relativizing its quantifiers to the definable class M is an ordinary formula of V . Separation in V and the substitution observation give Separation in M . If such a formula defines a function on a set of M with values in M , Replacement in V produces the range; its definition and its hereditary constituents put the range in M . Extensionality and Foundation follow by transitivity.

Finally, restrict the canonical $\text{OD}(q, p)$ well-order to a set $x \in M$. This restriction is definable from q, p and x , hence from q, p and ordinals. Its transitive closure consists of the relation itself, finite set constructions from elements of x , and their transitive closures. All these are in $\text{OD}(q, p)$. The restriction thus belongs to M and well-orders x there. This proves Choice. $\square$

We will not infer Choice for a larger hereditary class obtained by allowing *all* of V_λ as separate parameters at once. The fixed-parameter models in Lemma 2.4 suffice, and avoid that extra issue entirely.

3 One witness, uniformly correct for all parameters

3.1 The critical sequence

Lemma 3.1 (Critical-sequence normalization). Let $j : X \rightarrow V_\zeta$ witness ultraexactingness at a height $\zeta > \lambda + \omega$, and put

$$e = j \upharpoonright V_\lambda, \quad \kappa = \text{crit}(e), \quad \kappa_n = e^n(\kappa), \quad c = \{\kappa_n : n < \omega\}.$$

Then $e : V_\lambda \rightarrow V_\lambda$ is elementary,

$$\sup_{n < \omega} \kappa_n = \lambda, \quad e(\xi) > \xi \quad (\kappa \leq \xi < \lambda), \quad e \upharpoonright V_\kappa = \text{id}.$$

Moreover $c, q = [c] \in X$, and

$$j(c) = c \setminus \{\kappa\}, \quad j(q) = q.$$

Proof. The object V_λ is definable from λ at the chosen height, so belongs to X and is fixed by j . Relativizing any formula to V_λ , and using $V_\lambda \subseteq X$, shows that e is an elementary self-embedding. This reasoning does not require transitivity of X .

The critical point is below λ because $j \upharpoonright \lambda$ is nonidentity. Elementary embeddings are strictly increasing on ordinals and never move an ordinal downward. It follows that $\kappa < \kappa_1 < \kappa_2 < \dots < \lambda$. Suppose $\mu = \sup_n \kappa_n < \lambda$. The sequence then belongs to V_λ , and elementarity shows $e(\mu) = \mu$. The restriction to $V_{\mu+2}$ is a nonidentity elementary self-embedding, contrary to local Kunen. Here $\mu + 2 < \lambda$: λ is an infinite cardinal, whereas $\mu < \lambda$. Thus $\sup_n \kappa_n = \lambda$, and in particular λ is a limit cardinal of cofinality ω . For $\kappa_n \leq \xi < \kappa_{n+1}$ we have $e(\xi) \geq \kappa_{n+1} > \xi$.

An induction on rank proves $e \upharpoonright V_\kappa = \text{id}$. At a rank $\alpha < \kappa$, the ordinal α and V_α are fixed. If all members of V_α are already fixed, membership in any $x \subseteq V_\alpha$ is preserved pointwise, and $e(x)$ is still a subset of V_α . Hence $e(x) = x$.

Since the set e belongs to X , its first moved ordinal and its ω -iteration on that ordinal are definable in V_ζ from e . They belong to X by elementary definable closure. In particular $c \in X$. The increasing enumeration of c is in X as well. j fixes every natural number, so applying j to that enumeration shows $j(c) = j''c = \{\kappa_{n+1} : n < \omega\}$.

Finally q is definable from c, λ . All subsets of λ and their finite-modification classes occur below the chosen height, so this definition computes the actual class, not a truncated one. Elementarity gives $j(q) = [j(c)] = [c] = q$. $\square$

3.2 Uniform reflection is chosen before the witness

The argument must handle *all* ordinal definitions, including ones using ordinals above the critical point. It would be incorrect to choose an arbitrary definition of a counterexample and simply assert that its ordinal parameters are fixed.

For each of the finitely many counterexample properties used below, fix ordinary first-order formulas expressing membership in $\text{OD}(q, p)$, existence of a counterexample, its least permitted parameter rank, and selection of the canonical least counterexample. These are fixed formulas, even though they quantify over codes for arbitrarily many definitions.

By the reflection theorem choose, *before choosing j* , a cardinal $\zeta > \lambda + \omega$ such that V_ζ is correct for these formulas and their required subformulas, for *every* tuple of parameters in V_ζ . One may choose a sufficiently large finite n and take ζ in $C^{(n)}$ and in the club of cardinal closure points. This is finite-level correctness, not the assertion $V_\zeta < V$.

In this report the two counterexample predicates concern subsets of λ and functions $\lambda \rightarrow \lambda$. Such objects are all in $V_{\lambda+\omega}$; therefore the selected counterexamples lie below ζ . Choose the ultraexacting witness at this ζ , and form c, q as in Lemma 3.1.

Lemma 3.2 (Fixed-counterexample principle). Fix one of the preceding counterexample formulas $B(x, q, \lambda)$. If

$$(\exists p \in V_\lambda)(\exists x \in \text{OD}(q, p)) B(x, q, \lambda),$$

then there is such a counterexample $x \in X$ with $j(x) = x$.

Proof. Let $\rho < \lambda$ be least such that some $p \in V_\rho$ permits a counterexample in $\text{OD}(q, p)$. The minimum exists by the displayed hypothesis and the fact that λ is a limit ordinal. Its defining formula uses only q, λ . Uniform correctness ensures that V and V_ζ compute the same ρ . Since $q, \lambda \in X$ are fixed, elementary definable closure puts ρ in X , and elementarity gives $j(\rho) = \rho$. By Lemma 3.1, $\rho < \kappa$.

Choose any witnessing $p \in V_\rho$. This p belongs to $V_\kappa \subseteq X$ and is fixed by j . Among the counterexamples in $\text{OD}(q, p)$ choose the least under the canonical well-order of Lemma 2.4. Its selection is a unique definition from q, p, λ . The height was chosen correct for that selection formula, so the selected x is the same in V and V_ζ . Elementarity puts it in X and fixes it.

Notice that no ordinal parameter in the *original* definition of x was assumed fixed. The minimization and canonical selection replace that invalid assumption by a unique fixed definition. $\square$

The minimal-rank mechanism is inherited from the synthesis's cofinal-orbit-rigidity argument [1]. Its use for the two new counterexample properties is what permits a single tail class to work uniformly for all low-rank parameters.

4 The two tail theorems

Fix from now on the uniformly correct witness of Section 3, its critical sequence c , and $q = [c]$.

Theorem 4.1 (No relative-definable splitter). For every $A \subseteq \lambda$ with $A \in \mathcal{O}_q$, exactly one of

$$\text{Tail}(q, A), \quad \text{Tail}(q, \lambda \setminus A)$$

holds. Equivalently, every such A either contains almost all of c or meets c in only finitely many points.

Proof. If neither alternative holds, A splits q . Splitting is expressible using q, λ, A , without selecting a representative. Apply Lemma 3.2 to the property of being a splitting subset of λ . It gives a splitting $A \in X$ with $j(A) = A$. For every $n < \omega$, elementarity of j , and of the inclusion $X < V_\zeta$, gives

$$\kappa_n \in A \iff j(\kappa_n) \in j(A) \iff \kappa_{n+1} \in A.$$

Thus the membership pattern on c is constant, contradicting that A splits q . Both alternatives cannot hold because c is infinite. $\square$

Definition 4.2 (Eventual regressivity). A function $f : \lambda \rightarrow \lambda$ is *eventually regressive on q* if $f(\alpha) < \alpha$ for all but finitely many α in a representative of q . It is *eventually constant on q* if some $\xi < \lambda$ satisfies $\text{Tail}(q, \{\alpha : f(\alpha) = \xi\})$. Both definitions are independent of the representative.

Theorem 4.3 (Tail normality). Every $f : \lambda \rightarrow \lambda$ in $\mathcal{O}_q$ that is eventually regressive on q is eventually constant on q .

Proof. Suppose the theorem fails. Its failure is a first-order property $B(f, q, \lambda)$: f is a total function of the indicated type, is eventually regressive, and is not eventually constant. Lemma 3.2 supplies a counterexample $f \in X$ fixed by j .

For every n , $f(\kappa_n)$ belongs to X , because it is uniquely definable there from f, κ_n . It is below λ , so j agrees with e at that value. Consequently

$$f(\kappa_{n+1}) = e(f(\kappa_n)) \quad (n < \omega). \tag{4.1}$$

Moreover, elementarity gives the two-way equivalence

$$f(\kappa_n) < \kappa_n \iff f(\kappa_{n+1}) < \kappa_{n+1}.$$

Eventual regressivity therefore implies regressivity at *every* entry, in particular

$$\xi := f(\kappa_0) < \kappa_0 = \kappa.$$

This value is fixed by e . Equation (4.1), by ordinary induction on n , now yields $f(\kappa_n) = \xi$ for every n . This contradicts failure of eventual constancy. $\square$

Where normality enters

For subsets, shift invariance only says that a binary membership pattern is constant. For regressive functions, the first value lies *below the critical point*, hence is fixed. That extra fact forces all subsequent values to be the same. Starting with an arbitrary orbit above the critical point does not supply this fact.

Corollary 4.4 (No definable representative). No member of q belongs to O_q . In particular, $q \notin M_{q,p}$ and $q \cap M_{q,p} = \emptyset$ for all $p \in V_\lambda$.

Proof. If $a \in q \cap OD(q, p)$, let A consist of the even-indexed entries of its increasing enumeration. Then $A \in OD(q, p)$ and A splits q , contradicting Theorem 4.1. If $q \in M_{q,p}$, transitivity would place every member of q in $M_{q,p} \subseteq OD(q, p)$. $\square$

5 The internal normal-measure theorem

Theorem 5.1 (Canonical internal normal measures). Let λ be ultraexacting. There is $q \in Q_\lambda$ such that, for every $p \in V_\lambda$, the set

$$U_{q,p} = \{A \in \mathcal{P}(\lambda) \cap M_{q,p} : \text{Tail}(q, A)\} \quad (5.1)$$

belongs to $M_{q,p}$ and is a nonprincipal normal λ -complete ultrafilter on λ in that model. In particular

$$M_{q,p} \models \text{“}\lambda \text{ is measurable.”}$$

The same q can be used simultaneously for all $p \in V_\lambda$.

Proof. Take the q from Section 4. Fix $p \in V_\lambda$, and abbreviate $M = M_{q,p}$ and $U = U_{q,p}$.

The measure is an element of the model. The right side of (5.1) defines a set in V from q, p and the ordinal λ , using the definable predicate for M . Hence $U \in OD(q, p)$. Every member A of U is a subset of an ordinal and belongs to M . Its transitive closure is contained in $OD(q, p)$. Together with $U \in OD(q, p)$, this proves $\text{tc}(\{U\}) \subseteq OD(q, p)$, or $U \in M$. This is stronger than merely having an external ultrafilter on the Boolean algebra $\mathcal{P}(\lambda) \cap M$.

Ultrafilter and nonprincipality. The set λ belongs to U and the empty set does not. If two sets contain almost all of a representative of q , their intersection does too. Upward closure is immediate. For every $A \in \mathcal{P}(\lambda) \cap M$, Theorem 4.1 selects exactly one of A and $\lambda \setminus A$. These properties are the ultrafilter axioms for $\mathcal{P}^M(\lambda)$. Every final segment of λ is in U , since representatives are increasing cofinal ω -sets. No bounded set, and in particular no singleton, is in U .

Normality. Let $S \in U$ and let $f \in M$ be a regressive function on $S \setminus \{0\}$ with values below λ . Extend it to a total function $\tilde{f} : \lambda \rightarrow \lambda$ by setting it equal to zero outside $S \setminus \{0\}$. This total function is in M and

hence in $\text{OD}(q, p)$. Since a representative of q is eventually in $S \setminus \{0\}$, $\bar{f}$ is eventually regressive on q . Theorem 4.3 supplies $\xi < \lambda$ with $\text{Tail}(q, \{\alpha : \bar{f}(\alpha) = \xi\})$. Intersecting with $S \setminus \{0\}$ gives

$$\{\alpha \in S \setminus \{0\} : f(\alpha) = \xi\} \in U.$$

The fiber is in M by Separation. This proves the internal normality statement, with all its quantifiers interpreted in M .

λ -completeness. Let $\tau < \lambda$ and let $\langle A_i : i < \tau \rangle \in M$ be a sequence of members of U . For $\tau = 0$ the assertion is immediate. Otherwise suppose $A = \bigcap_{i < \tau} A_i \notin U$. Then $S = \lambda \setminus A \in U$. In M define

$$f(\alpha) = \min\{i < \tau : \alpha \notin A_i\} \quad (\alpha \in S).$$

On $S \setminus (\tau + 1) \in U$ this is regressive. Normality yields an $i < \tau$ such that

$$T = \{\alpha \in S \setminus (\tau + 1) : f(\alpha) = i\} \in U.$$

But $T \cap A_i = \emptyset$, contradicting the filter property. Thus the intersection belongs to U .

Finally λ remains an uncountable cardinal in M , since an inner-model bijection contradicting that assertion would be a bijection in V as well. The preceding properties therefore witness its measurability in M . Uniformity in p follows because the two tail theorems were proved for all of $\mathcal{O}_q$ using the same q . $\square$

Corollary 5.2 (Regularity and strong limit, internally). Each $M_{q,p}$ regards λ as strongly inaccessible.

Proof. We give the standard deductions to make the internal quantifiers explicit. Work entirely in $M = M_{q,p}$. A cofinal sequence $\langle \alpha_i : i < \tau \rangle$ with $\tau < \lambda$ would give the τ many members $\lambda \setminus (\alpha_i + 1)$ of U with empty intersection, contradicting completeness. Hence λ is regular.

Suppose $\eta < \lambda$ and $2^\eta \geq \lambda$. By Choice in M there is an injection $h : \lambda \rightarrow \mathcal{P}(\eta)$. For each $\xi < \eta$, let $B_\xi = \{\alpha < \lambda : \xi \in h(\alpha)\}$. Select the one of B_ξ and $\lambda \setminus B_\xi$ belonging to U . Their intersection is in U by completeness, and all its members have exactly the same h -value. Injectivity would make that intersection have at most one member, contradicting nonprincipality. Thus $2^\eta < \lambda$ for all $\eta < \lambda$. $\square$

Proposition 5.3 (Uniqueness and coherence). For the q of Theorem 5.1:

1. $U_{q,p}$ is the unique ultrafilter on $\mathcal{P}^{M_{q,p}}(\lambda)$ whose every member contains almost all of a representative of q .
2. If $\text{OD}(q, p) \subseteq \text{OD}(q, r)$, then $M_{q,p} \subseteq M_{q,r}$ and

$$U_{q,p} = U_{q,r} \cap \mathcal{P}^{M_{q,p}}(\lambda).$$

3. Given $p_0, p_1 \in V_\lambda$, coding the pair by $r \in V_\lambda$ gives a common larger model $M_{q,r}$ and coherent measures as in part (2).

Proof. For part (1), the assumed tail property makes every member of the other ultrafilter a member of $U_{q,p}$. Two ultrafilters on the same Boolean algebra cannot be properly included one in the other: the complement of any proposed missing member gives a contradiction. For part (2), the hereditary definitions give the model inclusion, and the same formula $\text{Tail}(q, A)$ decides both measures on their common subsets. Coding gives part (3). $\square$

5.1 Why ambient countable incompleteness is harmless

Let $c = \{\kappa_n : n < \omega\}$ be the chosen critical sequence. Every set $T_n = \lambda \setminus (\kappa_n + 1)$ is in $U_{q,p}$, yet in the ambient universe

$$\bigcap_{n < \omega} T_n = \emptyset. \quad (5.2)$$

This explicitly shows that $U_{q,p}$ is not externally countably complete. The sequence $\langle T_n : n < \omega \rangle$ is not an element of $M_{q,p}$: otherwise its internal completeness would contradict (5.2). Each individual T_n is in the model, but the external enumeration of them is not.

Nor does (5.2), by itself, prove that the ultrapower of $M_{q,p}$ using only its *internal* functions is externally ill-founded. An external sequence of sets and an internal sequence of functions are different data. We make no iterability or external ultrapower well-foundedness assertion.

6 Canonical Prikry layers

Definition 6.1 (Prikry forcing and the filter from a sequence). Let M be a transitive inner model of ZFC containing a normal measure U on its measurable cardinal λ . A condition in $\mathbb{P}_U^M$ is (s, A) , where s is a finite increasing sequence of ordinals below λ , $A \in U$, and every member of A is above every entry of s . The stronger-condition convention is used: $(t, B) \leq (s, A)$ if t end-extends s , its new entries belong to A , and $B \subseteq A$. A direct extension, written $\leq^*$, keeps the stem.

For $a \in D_\lambda$, identify a with its increasing enumeration and put

$$G_a = \{(s, A) \in \mathbb{P}_U^M : s \text{ is an initial segment of } a, a \setminus \text{ran}(s) \subseteq A\}.$$

Lemma 6.2 (Mathias's criterion). If $a \in D_\lambda$ satisfies $a \subseteq^* A$ for every $A \in U$, then G_a is $\mathbb{P}_U^M$ -generic over M .

A proof of this classical criterion, entirely with the internal sequences and dense sets of M , appears in Section A. This is the form needed here; no countability assumption on M is made.

Theorem 6.3 (The canonical layer). Let q be as in Theorem 5.1. For every $p \in V_\lambda$ and every $a \in q$, a is Prikry generic over $M_{q,p}$ for $U_{q,p}$. Define

$$K_{q,p} = M_{q,p}[G_a] = M_{q,p}[a].$$

Then $K_{q,p}$ is independent of the choice of $a \in q$, and

$$M_{q,p} \subsetneq K_{q,p} \subseteq V, \quad q \in K_{q,p}.$$

The two inner models have the same cardinals and

$$V_\lambda^{M_{q,p}} = V_\lambda^{K_{q,p}}, \quad \text{cf}^{K_{q,p}}(\lambda) = \omega, \quad (\lambda^+)^{M_{q,p}} = (\lambda^+)^{K_{q,p}}. \quad (6.1)$$

Proof. By the definition of $U_{q,p}$, every $A \in U_{q,p}$ contains almost all of *every* representative $a \in q$. Lemma 6.2 therefore applies. The resulting generic filter is a set of V , and evaluating names from $M_{q,p}$ produces a transitive inner model of V . It is a genuine set-forcing extension of $M_{q,p}$, and hence satisfies ZFC. The generic filter is definable from a and the forcing parameters, while a is recovered as the union of its stems. This explains $M_{q,p}[G_a] = M_{q,p}[a]$ without treating an arbitrary notation $M[a]$ as an independently given forcing extension.

If $a, b \in q$, the finite set $a \triangle b$ belongs to $M_{q,p}$, because every finite set of ordinals does. Thus a and b are mutually recoverable over $M_{q,p}$ and give the same extension. This proves that $K_{q,p}$ is well-defined. The full equivalence class q is in $K_{q,p}$: it consists precisely of the finite modifications of a , and all finite subsets of λ are already present. This calculation gives the actual ambient class, not merely a smaller class of representatives computed internally. Properness follows from Corollary 4.4.

The standard preservation facts for normal Prikry forcing are proved in Section A: there are no new bounded subsets of λ , the forcing is λ^+ -cc internally, and all cardinals are preserved. The sequence a changes the cofinality of λ to ω . These give the successor assertion.

For rank agreement, induct on $\alpha < \lambda$. In $M_{q,p}$, λ is strongly inaccessible by Corollary 5.2, so $|V_\alpha| < \lambda$. If the rank segments up to α agree, a new subset of V_α could be coded, using a bijection in $M_{q,p}$, by a new bounded subset of λ . There are no such subsets. Successor stages therefore agree, and limit stages are unions. This proves (6.1). $\square$

Remark 6.4 (A layer is not an ambient ground). The theorem proves that $M_{q,p}$ is a ground of $K_{q,p}$. It does not prove that either model is a ground of V , or that $K_{q,p} = V$. No quotient forcing from $K_{q,p}$ to V is supplied. This distinction is essential in the stationary-set application below.

Corollary 6.5 (The coded boundary). Assume additionally $V_\lambda \subseteq \text{HOD}^V$. Then $M_{q,p} = M_q$ and $U_{q,p} = U_q$ for every $p \in V_\lambda$. The common Prikry layer K_q satisfies

$$V_\lambda^{M_q} = V_\lambda^{K_q} = V_\lambda^V, \quad M_q \models \text{“}\lambda \text{ is measurable.”}$$

Nevertheless no representative of q belongs to M_q .

Proof. Each $p \in V_\lambda$ is ordinal definable in V , so it can be eliminated from any $\text{OD}(q, p)$ definition by substitution. Hence $\text{OD}(q, p) = \text{OD}(q)$ and the hereditary models are equal. Every element of V_λ and its transitive closure are in $\text{HOD} \subseteq M_q$, so $V_\lambda \subseteq M_q$. Transitivity and inclusion in V give the rank equalities. The remaining statements were proved above. $\square$

The measurable cardinal here is λ in M_q , not in ordinary HOD and not in V . There is no argument eliminating the high-rank parameter q from the definition of U_q .

7 Sharpness and diagnostic counterexamples

7.1 The low-rank parameter bound cannot be raised

Allow the additional parameter $p = c$, which belongs to $V_{\lambda+1}$ but not V_λ . The subset

$$A = \{\kappa_{2n} : n < \omega\}$$

is definable from c and splits q . Also the function which assigns to α the largest member of $c \cap \alpha$ when that set is nonempty, and assigns zero otherwise, is definable from c . At κ_n for $n > 0$ it takes the value κ_{n-1} . It is eventually regressive and not eventually constant. Thus both tail theorems genuinely fail when parameters from $V_{\lambda+1}$ are allowed indiscriminately.

7.2 A fixed recoding with a nonnormal tail ultrafilter

Let

$$c^+ = \{\kappa_n + 1 : n < \omega\}, \quad q^+ = [c^+].$$

The ordinal map $h(\alpha) = \alpha + 1$ sends λ into λ . The class q^+ is definable from q , without selecting a representative: apply h to any representative and take its finite-difference class. Conversely, apply predecessor to the successor ordinals in a representative of q^+ and ignore the finitely many nonsuccessors. The resulting class is q . Both operations are well-defined modulo finite difference.

Consequently

$$\text{OD}(q, p) = \text{OD}(q^+, p), \quad M_{q,p} = M_{q^+,p}. \quad (7.1)$$

Also $q^+ \in X$ and $j(q^+) = q^+$, since the recoding is definable from fixed parameters. The no-splitting property transfers: if A split q^+ , then $h^{-1}[A]$ would split q and would be definable from the same parameters. Thus the tail rule on q^+ is an ultrafilter on the same internal Boolean algebra.

Proposition 7.1 (Normality distinguishes the critical class). In the common model $M = M_{q,p} = M_{q^+,p}$, the tail ultrafilter

$$U^+ = \{A \in \mathcal{P}^M(\lambda) : \text{Tail}(q^+, A)\}$$

is λ -complete and nonprincipal but is not normal.

Proof. The recoding gives

$$A \in U^+ \iff h^{-1}[A] \in U_{q,p}.$$

Preimages commute with intersections, so internal λ -completeness follows. The formula also shows that U^+ is an element of M ; alternatively use its external definition and the same hereditary argument as for $U_{q,p}$. The set S of successor ordinals below λ belongs to U^+ . On S the function $f(\alpha + 1) = \alpha$ is an ordinal-definable regressive function. Every fiber is a singleton, hence is outside U^+ . This contradicts the normality condition. $\square$

This example pinpoints a possible false strengthening: neither $j(q) = q$ nor equality of the relative HOD models forces normality of *every* tail presentation. In the proof of Theorem 4.3, the inequality needed is $f(\kappa) < \kappa$, not merely $f(\beta) < \beta$ at an arbitrary orbit root $\beta > \kappa$.

7.3 What no-splitting does not assert

Choice still supplies splitting subsets for all $q \in Q_\lambda$. Our obstruction concerns their restricted definability, not their existence as sets. The tail measure need not decide arbitrary ambient subsets of λ : the even and odd pieces of c are already counterexamples. Also the normal measures belonging to different fixed-parameter models need not form a single internally indexed sequence in any one of them. The coherence of Proposition 5.3 is not a claim of such internal uniform enumeration.

8 Conditional inconsistency and equiconsistency

8.1 A splitting principle incompatible with ultraexactingness

Definition 8.1 (Relative-definable splitting). For an uncountable cardinal λ of cofinality ω , let $\text{DSplit}(\lambda)$ be the assertion

$$\forall q \in Q_\lambda \exists p \in V_\lambda \exists A \subseteq \lambda (A \in \text{OD}(q, p) \wedge A \text{ splits } q).$$

This is a pointwise definability principle; it does not demand a uniform choice of the definitions or their parameters.

Theorem 8.2 (Conditional inconsistency). The following theory is inconsistent:

$$\text{ZFC} + \exists \lambda (\text{UEx}(\lambda) \wedge \text{DSplit}(\lambda)).$$

In particular, at an ultraextending λ there is no OD_{V_λ} function $F : Q_\lambda \rightarrow \mathcal{P}(\lambda)$ such that $F(q)$ splits q for every $q \in Q_\lambda$.

Proof. Take the critical tail class supplied by Theorem 4.1. It has no splitter in $\text{OD}(q, p)$ for any $p \in V_\lambda$, contradicting $\text{DSplit}(\lambda)$. If F were ordinal definable from a low-rank parameter p , its value at q would be in $\text{OD}(q, p)$ by substitution, giving the same contradiction. $\square$

This is not a failure of Choice. Since Q_λ is a set, Choice selects a representative for every class. Taking the even entries of each selected representative gives an unrestricted splitting function. Thus the inconsistency theorem isolates the definability restriction, rather than disguising an impossible demand for ordinary splitting subsets.

8.2 The new structural package

Let $\text{NMP}(\lambda)$ abbreviate the following assertion. There is $q \in Q_\lambda$ such that, for every $p \in V_\lambda$, the class $M_{q,p}$ of Definition 2.3 contains the normal measure $U_{q,p}$ of (5.1), and every $a \in q$ generates a $\mathbb{P}_{U_{q,p}}^{M_{q,p}}$ -generic filter over that class. The resulting $K_{q,p}$ is independent of a and has the preservation properties of Theorem 6.3.

This notation is not a new second-order axiom. Membership in $M_{q,p}$ is expressed by a fixed first-order formula. Normality and completeness are relativizations to that formula. Genericity says that the set G_a meets every dense subset belonging to the class; that too is a first-order assertion. Name evaluation defines the corresponding layer. Statements that these classes satisfy individual axioms of ZFC follow by the arguments already given, rather than by quantifying over a truth predicate for a proper class.

Theorem 8.3 (Intrinsic structure).

$$\text{ZFC} \vdash \forall \lambda (\text{UEx}(\lambda) \longrightarrow \text{NMP}(\lambda)).$$

With $V_\lambda \subseteq \text{HOD}$ added, one can in addition require that all the fixed-parameter models and measures coincide and that their rank segments below λ equal those of the ambient universe.

Proof. Apply Theorems 5.1 and 6.3, followed in the coded case by Corollary 6.5. $\square$

Here I0 means a nontrivial elementary embedding $i : L(V_{\eta+1}) \rightarrow L(V_{\eta+1})$ with critical point below η . The comparison does not identify η with the λ occurring in a different consistency model.

Theorem 8.4 (Enhanced I0 calibration). The following theories are equiconsistent:

$$\begin{aligned} T_0 &= \text{ZFC} + \text{“There is an I0 embedding”}, \\ T_{\text{layer}} &= \text{ZFC} + \exists \lambda (\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD} \wedge \text{NMP}(\lambda)). \end{aligned}$$

In the second theory one may also include the full low-rank agreement and common-model assertions of Corollary 6.5, and $\neg \text{DSplit}(\lambda)$, without changing the conclusion.

Proof. Forward relative consistency. The imported comparison theorem gives

$$\text{Con}(T_0) \implies \text{Con}(\text{ZFC} + \exists \lambda \text{ UEx}(\lambda)).$$

In an ultraexacting model use the imported coding forcing of [4, Proposition 3.9] to retain ultraexactingness of λ and obtain $V_\lambda \subseteq \text{HOD}$. This is a relative-consistency construction, so it gives the corresponding implication between formal consistency statements, not an assumption that the universe already has an external generic filter.

In the resulting model, apply the internal ZFC proof of Theorem 8.3. It supplies $\text{NMP}(\lambda)$, the common model and full rank agreement. Theorem 8.2 supplies $\neg \text{DSplit}(\lambda)$. Thus the enhanced theory is relatively consistent.

Reverse relative consistency. Forget all the extra assertions of T_{layer} and retain an ultraexacting cardinal. The reverse direction of the imported comparison theorem then gives $\text{Con}(T_0)$. $\square$

What the calibration does and does not establish

The additional structure proved here costs no consistency strength *over the already calibrated ultraexacting boundary theory*. This is not a new proof of the ultraexacting–I0 comparison and not a consistency-strength separation. Conversely, the proof supplies more than an example in a specially chosen Prikry extension: Theorem 8.3 constructs a canonical relative-HOD layer in *every* universe with an ultraexacting cardinal.

9 The stationary-set gap above the canonical layer

This section makes the archive’s successor obstruction concrete for the new models. The general collapse-or-destroy argument was already present in the synthesis [1, the successor-stationarity theorem]; the canonical Prikry layer, and its exact role in the application, are the additional structure here.

Fix q, p , and abbreviate

$$M = M_{q,p}, \quad K = K_{q,p}, \quad \theta = (\lambda^+)^M = (\lambda^+)^K, \quad S = (E_\lambda^\theta)^M.$$

Thus S is the set of ordinals below θ whose cofinality is λ in M .

Lemma 9.1 (The stationary witness). S is stationary in θ in K , and $K \models S \subseteq E_\omega^\theta$. In every outer model $W \supseteq K$ of ZFC, S fails to reflect at every $\beta < \theta$ with $\text{cf}^W(\beta) > \omega$.

Proof. In M , λ is regular and $\theta = \lambda^+$ is regular, so E_λ^θ is stationary. Explicitly, inside any club in θ construct an increasing continuous sequence of length λ ; its supremum is below θ , is in the club, and has cofinality λ . Normal Prikry forcing is θ -cc in M , so it preserves this stationary set. A proof of this chain-condition preservation fact is included below in Lemma A.6.

For each $\alpha \in S$, M has an increasing cofinal map $h_\alpha : \lambda \rightarrow \alpha$. Composing it with a cofinal ω -sequence in λ belonging to K gives a cofinal ω -sequence in α . Thus $\text{cf}^K(\alpha) = \omega$, and the same is true in every outer W .

Now fix $\beta < \theta$ with $\text{cf}^W(\beta) > \omega$. Let $\mu = \text{cf}^M(\beta)$. Since $\beta < \lambda^+$ in M , $\mu \leq \lambda$. If $\mu = \lambda$, composition as above would give $\text{cf}^W(\beta) = \omega$, impossible. If $\mu \leq \omega$, the cofinal sequence in M would already contradict $\text{cf}^W(\beta) > \omega$. Consequently $\omega < \mu < \lambda$.

In M choose a strictly increasing continuous cofinal map $h : \mu \rightarrow \beta$. The set

$$C = \{h(\xi) : \xi < \mu \text{ is a nonzero limit ordinal}\}$$

is club in β . At each such ξ , continuity gives $\text{cf}^M(h(\xi)) \leq \text{cf}^M(\xi) < \lambda$; hence $C \cap S = \emptyset$. Being closed and unbounded in a fixed ordinal is absolute between transitive models containing the set, so this same C is a club witness in W . Therefore $S \cap \beta$ is nonstationary in β . $\square$

For completeness we give the reflection fact used in the application. Use the usual ultrafilter formulation of strong compactness: for every cardinal $\tau \geq \delta$ there is a fine δ -complete ultrafilter on $P_\delta(\tau)$.

Lemma 9.2 (Strong compactness reflects countable cofinality). If δ is strongly compact and $\theta > \delta$ is regular, every stationary $T \subseteq E_\omega^\theta$ reflects at some $\beta < \theta$ with $\omega < \text{cf}(\beta) < \delta$.

Proof. Take the well-founded ultrapower $i : V \rightarrow N$ by a fine δ -complete ultrafilter on $P_\delta(\theta)$. Its seed gives a set $t \in N$ such that $i''\theta \subseteq t \subseteq i(\theta)$ and $|t|^N < i(\delta)$. Put $\gamma = \sup i''\theta$. Since $i(\theta)$ is regular in N and exceeds $i(\delta)$, $\gamma < i(\theta)$; also $\text{cf}^N(\gamma) < i(\delta)$. Externally $\text{cf}(\gamma) = \theta$: an increasing cofinal sequence of length θ is supplied by $i''\theta$, and a shorter cofinal set would pull back to an unbounded subset of the regular θ of smaller size. In particular $\text{cf}^N(\gamma) > \omega$.

We claim $i(T) \cap \gamma$ is stationary in γ in N . Let $C \in N$ be any club in γ . For each $\alpha < \theta$, choose $g(\alpha) > \alpha$ such that some point of C lies strictly between $i(\alpha)$ and $i(g(\alpha))$. This is possible because both C and $i''\theta$ are cofinal in γ . Let $D \subseteq \theta$ be the club of limit ordinals closed under g . Pick $\alpha \in D \cap T$. Since $\text{cf}(\alpha) = \omega$ and $\text{crit}(i) = \delta > \omega$, i is continuous at α . Choose $\alpha_n \nearrow \alpha$; there are points $d_n \in C$ with

$$i(\alpha_n) < d_n < i(g(\alpha_n)) < i(\alpha).$$

Their supremum is $i(\alpha)$, so closure of C gives $i(\alpha) \in C \cap i(T) \cap \gamma$. This proves the claim for every club C in N .

In N , therefore, $\gamma < i(\theta)$ witnesses reflection of $i(T)$ at an ordinal of uncountable cofinality below $i(\delta)$. Elementarity gives the claimed reflection point below θ in V . $\square$

Theorem 9.3 (A canonical collapse-or-destroy alternative). If the ambient universe V has a strongly compact cardinal $\delta < \lambda$, then exactly one of the following cardinal cases holds, with the indicated additional conclusion in the second case:

1. $|\theta|^V = \lambda$; or
2. $\theta = (\lambda^+)^V$ and S is nonstationary in V .

In particular $K \neq V$. There is no representation $V = K[H]$ by a forcing that is θ -cc in K , and no such representation by a forcing with a dense subset of K -cardinality below θ .

Proof. Every ordinal in $[\lambda, \theta)$ has M -cardinality λ , so has V -cardinality λ as well. Since λ is still a cardinal in V , either θ also has V -cardinality λ , or it is precisely $(\lambda^+)^V$. This gives the two cardinal cases.

In the latter case, suppose S remained stationary. The cofinal maps in Lemma 9.1 show $S \subseteq E_\omega^\theta$ in V . Lemma 9.2 would make it reflect at an ordinal of uncountable cofinality. This contradicts the final assertion of Lemma 9.1. Therefore S is nonstationary.

In K , θ is still a cardinal and S is still stationary; hence $K \neq V$. A θ -cc forcing over K preserves both those facts, again contradicting the alternative. A dense subset of size below θ implies the θ chain condition, proving the last assertion. $\square$

This theorem does not refute an ultraexacting cardinal above a strongly compact cardinal. It locates what the ambient universe must do *beyond the canonical Prikry layer*. The same argument works with the relevant stationary-reflection principle in place of strong compactness. No consistency of the combined large-cardinal hypothesis is being assumed or established here.

10 Audit, limitations, and next mathematical questions

10.1 Proof audit

The central implications use two counterexample predicates, not a scheme of individually chosen embeddings. Uniform correctness is fixed first; the critical tail class is then extracted from one witness; only afterward are all low-rank parameters quantified over. The essential checks are the following.

Potential failure	Reason it does not occur here
A moved ordinal parameter is silently fixed	The minimum parameter rank and the canonical counterexample are uniquely defined from fixed parameters; the original ordinal code is never assumed fixed.
The embedding domain is treated as transitive	Only $V_\lambda \subseteq X$ and elementary definable closure are used. The calculation on a countable set uses its enumeration in X .
A representative is admitted as a parameter	q is a single set parameter, not a list of permitted representatives. The separate bound is $p \in V_\lambda$.
The measure is merely external	Equation (5.1) and hereditary definability explicitly prove $U_{q,p} \in M_{q,p}$.
Internal completeness is confused with ambient completeness	Only sequences belonging to $M_{q,p}$ are used in the completeness proof. Equation (5.2) exhibits external failure.
A shifted orbit is assumed normal	Proposition 7.1 gives a concrete counterexample and isolates the special role of κ_0 .
Prikry genericity is asserted from cofinality alone	The tail condition is checked for every internal measure-one set, and the dense-set proof of Mathias's criterion is supplied.
An inner layer is called a ground of V	Only $M_{q,p} \subseteq K_{q,p}$ is proved to be a forcing-ground relation. The further inclusion into V is not assigned a forcing.
The I0 calibration is presented as new	The comparison and coding inputs are identified separately; only the enlarged canonical structure is proved here.

10.2 Questions not resolved by these proofs

Question 10.1. Can ordinary exactingness produce a tail class satisfying the two tail theorems, or an internal normal measure in some comparably canonical relative-definability model?

The present proof uses $e \in X$ to internalize the critical sequence and fix its full tail class. Removing that step would require a new argument, not merely deletion of “ultra” from the hypotheses.

Question 10.2. Under additional hypotheses, can one identify the relation between $K_{q,p}$ and V more precisely, for example as a specified forcing extension or by a covering/approximation property?

Theorem 9.3 gives an obstruction to small-chain-condition representations in the presence of a lower strongly compact cardinal. It gives no positive quotient forcing.

Question 10.3. How many different normal measures or relative HOD models arise from critical tail classes of sufficiently correct witnesses?

The archive's many rigid classes do not settle this: recodings can produce the same model, and even change a normal tail presentation into a nonnormal one. Counting classes, models, and normal measures are three distinct problems.

Question 10.4. What additional hypothesis can eliminate the parameter q from the measure, so that a normal measure on λ belongs to ordinary HOD?

Neither $V_\lambda \subseteq \text{HOD}$ nor the present canonicity argument eliminates q . It is important not to claim ordinary-HOD measurability from Corollary 6.5.

10.3 Verification and source scope

The supplied TeX synthesis and the Lean source interfaces were used as the baseline. No new Lean file was compiled and no machine verification of the new results is claimed. The ultraexacting witness interface and the comparison and coding results were checked in the primary arXiv versions cited below; Mathias's criterion was checked in Benhamou's treatment. Local Kunen remains the standard imported inconsistency theorem; its original proof is not reproduced or newly verified here. The literature check was targeted, not an exhaustive priority search. In particular, the established critical-sequence/ Prikry connection should not be mistaken for a novelty claim.

The complete paper proof supports the internal normal-measure theorem, its canonical-layer corollary, and the stated conditional consistency and inconsistency assertions. Specialist review should focus first on the uniform reflection convention and on the use of *hereditary* ordinal definability with a single high-rank set parameter.

A Prikry genericity and preservation: proofs

This appendix gives the classical facts used above, with all constructions performed inside a transitive model $M \models \text{ZFC}$. Within this appendix temporarily write ν for its measurable cardinal and $U \in M$ for its nonprincipal normal ν -complete ultrafilter. All sequences of sets selected in the lemmas belong to M . A reader can then take $\nu = \lambda$ and $M = M_{q,p}$. The criterion is due to Mathias [6]; [7, Section 3] is a further reference for these classical Prikry facts. The arguments below are included to expose the exact hypotheses needed in the application.

For a finite increasing sequence s , write $A_{>s} = \{\alpha \in A : (\forall \xi \in \text{ran}(s)) \xi < \alpha\}$. For the empty sequence, $A_{>s} = A$.

A.1 Normality gives diagonal thinning and homogeneity

Lemma A.1 (Finite-stem diagonal intersection). If $\langle A_s : s \in [\nu]^{<\omega} \rangle \in M$ and every A_s is in U , then

$$B = \{\beta < \nu : \forall s \in [\beta]^{<\omega} \beta \in A_s\}$$

belongs to U .

Proof. Suppose its complement belongs to U . Intersect that complement with $A_\emptyset$ and remove zero. For each remaining β , choose a nonempty finite $s_\beta \subseteq \beta$ such that $\beta \notin A_{s_\beta}$. Choice in M makes this an

internal function. The function $\beta \mapsto \max(s_\beta)$ is regressive. Normality gives a set $T \in U$ and $\alpha < \nu$ such that $\max(s_\beta) = \alpha$ for every $\beta \in T$.

There are fewer than ν finite subsets of $\alpha + 1$. By ν -completeness,

$$T \cap \bigcap \{A_s : s \in [\alpha + 1]^{<\omega}\} \in U.$$

Any β in that intersection must belong to A_{s_β} , contradicting its selection. Thus $B \in U$. $\square$

Lemma A.2 (Finite-arity homogeneity). If $r < \nu$ and $F : [\nu]^{<\omega} \rightarrow r$ belongs to M , there is $H \in U$ such that F is constant on $[H]^n$ for each fixed $n < \omega$. The constant may depend on n .

Proof. First prove the assertion for a single arity by induction. Arity zero is trivial. At arity one, exactly one of the fewer than ν color classes belongs to U : if none did, intersecting their complements would contradict ν -completeness.

Suppose the result holds for arity n and consider $F : [\nu]^{n+1} \rightarrow r$. For each increasing n -tuple s , choose a color $i_s < r$ such that

$$A_s = \{\beta > \max(s) : F(s \smallfrown \langle \beta \rangle) = i_s\} \in U.$$

For $n = 0$, interpret the bound on β as empty. The color exists by completeness and the ultrafilter property. By the induction hypothesis, there is $H_0 \in U$ on whose n -tuples s the value i_s is constant. Apply Lemma A.1 to the A_s , assigning ν to unused arities, and intersect the resulting set with H_0 . For any $(n + 1)$ -tuple from this intersection, its last entry is in the A_s indexed by the preceding entries. Its color is therefore the common i_s . This completes the induction.

For a coloring of all finite arities, select a homogeneous set for each n and intersect them. This is an internal countable sequence; $\omega < \nu$, so ν -completeness preserves membership in U . $\square$

A.2 The strong Prikry lemma

Lemma A.3 (Strong Prikry lemma). Let $D \in M$ be dense open in $\mathbb{P}_U^M$. For each condition (s, A) there are $A^* \subseteq A$ in U and $n < \omega$ such that

$$(s \smallfrown t, A_{>s \smallfrown t}^*) \in D \quad \text{for every } t \in [A^*]^n.$$

Proof. For each $t \in [A]^{<\omega}$, color t by one if some condition with stem $s \smallfrown t$ extending (s, A) belongs to D , and by zero otherwise. When its color is one, choose $B_t \in U$ witnessing

$$(s \smallfrown t, B_t) \leq (s, A), \quad (s \smallfrown t, B_t) \in D.$$

For other indices assign $B_t = \nu$; also assign ν to finite sequences outside A .

Extend the coloring arbitrarily to finite sequences outside A . Choose $H \subseteq A$ in U homogeneous at each arity for the coloring, using Lemma A.2 and then intersecting with A . Let $B \in U$ be the finite-stem diagonal intersection of all the B_t , and put $A^* = H \cap B$. Whenever t has color one, $A_{>s \smallfrown t}^* \subseteq B_t$, so openness of D gives $(s \smallfrown t, A_{>s \smallfrown t}^*) \in D$.

By density, some extension of (s, A^*) is in D . Its additional stem is a finite $t \subseteq A^*$, of some length n , and its color is one. Homogeneity at arity n makes every n -tuple from A^* have color one. This is the desired conclusion, including the possible case $n = 0$. $\square$

A.3 Proof of Mathias's criterion

Let $a = \langle \alpha_n : n < \omega \rangle$ be strictly increasing and cofinal in ν , possibly external to M , and assume it is eventually in each member of U . Form G_a as in Section 6.

First G_a is a filter. It is closed toward weaker conditions by the extension definition. Two stems of conditions in G_a are initial segments of the same sequence, hence comparable. Take the longer stem and intersect their measure sets, deleting entries below that stem. This gives a common stronger condition still in G_a .

Fix a dense open $D \in M$. For each finite stem s , apply Lemma A.3 below $(s, \nu_{>s})$, and choose in M a set $A_s \in U$ and a number n_s such that

$$(s \smallfrown t, (A_s)_{>s \smallfrown t}) \in D \quad (t \in [A_s]^{n_s}).$$

Let $B \in U$ be their finite-stem diagonal intersection. The sequence a is eventually in B . Choose an initial segment s of a containing all entries before that point. Every subsequent entry β of a is in B and is greater than all entries of s , so the diagonal property gives $\beta \in A_s$.

Let t be the next n_s entries of a . The condition $(s \smallfrown t, (A_s)_{>s \smallfrown t})$ is in D . Its stem is an initial segment of a and all remaining entries are in its measure set, so it is also in G_a . Hence G_a meets every dense open set of M , which proves Lemma 6.2. Arbitrary dense sets can be replaced by their downward closures; a filter meeting that closure also meets the original dense set by upward closure.

Conversely a Prikry-generic sequence is eventually in every $A \in U$, since conditions whose measure part is contained in A are dense. Conditions whose stem has an entry above any prescribed bound are also dense, as are conditions with stem length at least any prescribed natural number. These verify the familiar necessary cofinality and tail properties.

A.4 The ordinary Prikry property and bounded subsets

Lemma A.4 (Prikry property). Every forcing statement is decided by a direct extension of any given condition.

Proof. Fix a statement φ and a condition (s, A) . For each finite t from A , assign color zero if some direct extension with stem $s \smallfrown t$ forces φ , color one if one forces $\neg\varphi$, and color two if neither exists. Colors zero and one cannot both be possible, because conditions with the same stem have a common direct extension by intersecting their measure sets.

For each tuple with color zero or one choose a witnessing measure set B_t . Apply finite-arity homogeneity and diagonal thinning, exactly as in Lemma A.3, to obtain $A^* \subseteq A$ in U such that every tuple with color zero or one has its canonical condition $(s \smallfrown t, A^*_{>s \smallfrown t})$ forcing the assigned truth value, and the color is constant at each length.

There is a deciding extension of (s, A^*) . If its additional stem has length n , the common color at length n is zero or one. Every condition below (s, A^*) can be further extended to have at least n additional entries. The resulting condition extends the canonical condition associated with its first n new entries, so it forces the same truth value. Thus a dense set below (s, A^*) forces that value, and (s, A^*) itself forces it. $\square$

Corollary A.5 (No new bounded subsets). A $\mathbb{P}_U^M$ -generic extension has no new subset of any $\eta < \nu$.

Proof. Given a name for a subset of η and an initial condition, use the Prikry property successively for the statements that each $\xi < \eta$ belongs to the named set. Keep the stem unchanged. At limit stages intersect the measure sets chosen so far, and at the end intersect all of them. The recursion has length $\eta < \nu$, belongs to M , and ν -completeness ensures each intersection remains in U . The final direct extension decides membership for all $\xi < \eta$. The set of the affirmative decisions is a set of M . Such conditions are dense, proving the assertion. $\square$

A.5 Chain condition, stationary sets, and cardinals

Lemma A.6 (Chain-condition facts). In M , $\mathbb{P}_U$ is ν^+ -cc. More generally a θ -cc forcing, where θ is regular uncountable, preserves stationary subsets of θ and all cardinals at least θ .

Proof. For Prikry forcing, there are only ν finite stems. Any two conditions with the same stem are compatible, by intersecting their measure sets. An antichain therefore has size at most ν .

For stationarity, let $\mathbb{P}$ be θ -cc and let a condition u force that $\dot{C}$ is club in θ . For each $\alpha < \theta$, choose a maximal antichain below u deciding a point of $\dot{C}$ strictly above α . There are fewer than θ possible chosen points. By regularity their supremum is below θ ; choose a strict upper bound $b(\alpha) < \theta$. The set of limit ordinals $\gamma < \theta$ closed under b is a ground-model club D . For any $\gamma \in D$ and any $\alpha < \gamma$, the antichain ensures that u forces a point of $\dot{C}$ between α and γ . Hence u forces $\gamma \in \dot{C}$ by closure. A ground-model stationary set meets D , so no condition can force that it misses $\dot{C}$.

For cardinal preservation, a name for an ordinal value has fewer than θ possible values once a maximal deciding antichain is chosen. Therefore a name for a function with domain a cardinal $\mu < \xi$, where $\xi \geq \theta$ is a ground-model cardinal, cannot surject onto ξ . If $\xi = \theta$, regularity bounds the union of $\mu < \theta$ many sets of possible values below size θ . If $\xi > \theta$, that union has size at most $\mu \cdot \theta < \xi$. This excludes collapse of any such cardinal. The same argument bounds the possible range of a short alleged cofinal sequence in θ , so θ stays regular as used in the stationarity assertion. $\square$

Corollary A.7 (All-cardinal preservation). Normal Prikry forcing preserves every cardinal of M , changes $\text{cf}(\nu)$ to ω , and preserves $(\nu^+)^M$.

Proof. Cardinals at least ν^+ are preserved by Lemma A.6. Since ν is strongly inaccessible in M , functions between ordinals below ν can be coded by bounded subsets of ν . Corollary A.5 therefore excludes collapse of any cardinal below ν .

If ν itself collapsed to some $\mu < \nu$, an injection from ν into μ would restrict to an injection from a ground-model cardinal ρ with $\mu < \rho < \nu$ into μ . That restriction is coded below ν and so would already belong to M , a contradiction. Such a ρ exists because ν is a limit cardinal. Finally the generic union of stems has order type ω and is cofinal in ν , by the dense sets described above. $\square$

All the completeness and homogeneity arguments in this appendix were internal to M . The only external object was the candidate cofinal sequence in Mathias's criterion, and it was used *after* the internal dense-set constructions. This is why the appendix applies even though the tail measure in the ambient universe is explicitly countably incomplete.

References

- [1] *Large cardinals at the inconsistency frontier: a synthesis*. Second edition, 18 September 2026. Supplied in Cardinals3.zip, docs/Large_Cardinals_Synthesis.tex. In particular the critical-sequence normalization, minimal-rank-parameter device, cofinal-orbit rigidity, coded boundary package, and successor-stationarity obstruction. Archive source, not a publication or independent verification.
- [2] *Large cardinals at the inconsistency frontier: a unified assessment*. 18 September 2026. Supplied in Cardinals3.zip, docs/Large_Cardinals_Unified_Report.tex.
- [3] J. P. Aguilera, J. Bagaria, P. Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. [arXiv:2411.11568v4](#), revised 16 September 2025. Definition 3.3 and Lemma 3.2 for the ultraexacting witness interface. Primary version checked 18 September 2026.
- [4] J. P. Aguilera, J. Bagaria, G. Goldberg, P. Lücke. *Large cardinals beyond HOD*. [arXiv:2509.10254v1](#), 12 September 2025. Theorem A (equiconsistency), Proposition 3.9 (coding), and Section 5.2 (comparison with an iteration-based Prikry construction). Primary version checked 18 September 2026.
- [5] K. Kunen. Elementary embeddings and infinitary combinatorics. *Journal of Symbolic Logic* **36**(3) (1971), 407–413. [doi:10.2307/2269948](#). The local Kunen theorem is the standard imported inconsistency input, as in the supplied synthesis.
- [6] A. R. D. Mathias. On sequences generic in the sense of Prikry. *Journal of the Australian Mathematical Society* **15**(4) (1973), 409–414. [doi:10.1017/S1446788700028755](#). Attribution of the genericity criterion; a proof is included in this report.
- [7] T. Benhamou. *Prikry forcing and tree Prikry forcing of various filters*. [arXiv:1801.04424v2](#), revised 25 September 2018. Journal version: [doi:10.1007/s00153-019-00660-3](#). Section 3, especially Lemma 3.16 and Theorem 3.17. Version dates follow the arXiv submission record, not the date inserted by an experimental HTML rendering.